SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: T101 ARTIFICIAL INTELLIGENCE Practical

PRACTICAL - 7

AIM :- To implement the AdaBoost ensemble learning algorithm to predict whether a job applicant is suitable for selection based on Aptitude Score and Interview Score.

Dataset: applicant_selection.csv

```
Aptitude,Interview,Class
85,82,Selected
78,75,Selected
92,88,Selected
60,55,Not Selected
72,68,Not Selected
88,84,Selected
65,60,Not Selected
95,90,Selected
70,65,Not Selected
82,78,Selected
58,52,Not Selected
90,85,Selected
76,72,Selected
62,58,Not Selected
86,80,Selected
68,63,Not Selected
80,77,Selected
55,50,Not Selected
91,89,Selected
74,69,Not Selected
```

Code:- AdaBoost for Applicant Selection

```
#T101 RAJDEEP M PARAB
import pandas as pd
import warnings
warnings.filterwarnings("ignore")
from sklearn.metrics import confusion_matrix, accuracy_score, classification_report
# Reading the applicant dataset
data = pd.read_csv(
    r"C:\Users\rajde\OneDrive\Rajdeep's File\AI\AI_Practical-7\applicant_selection.csv"
)
print(data)
# Splitting the dataset into training and testing data
from sklearn.model_selection import train_test_split
training_set, test_set = train_test_split(
    data,
    test_size=0.2,
    random_state=1
)
```

```
# Selecting input features
X_train = training_set.iloc[:, 0:2].values
# Selecting output/class
Y_train = training_set.iloc[:, 2].values
X_test = test_set.iloc[:, 0:2].values
Y_test = test_set.iloc[:, 2].values
# Importing AdaBoost
from sklearn.ensemble import AdaBoostClassifier
# Creating AdaBoost model
adaboost = AdaBoostClassifier(
   n_estimators=100,
   learning_rate=1,
   random_state=1
)
# Training the model
adaboost.fit(X_train, Y_train)
# Predicting test data
Y_pred = adaboost.predict(X_test)
# Adding predictions to test dataset
test_set["Predictions"] = Y_pred
# Creating confusion matrix
results = confusion_matrix(Y_test, Y_pred)
print("Confusion matrix\n", results)
# Calculating accuracy
print("Accuracy score: ", accuracy_score(Y_test, Y_pred))
# Displaying classification report
print("Report: ")
print(classification_report(Y_test, Y_pred))
```

Output:

```
IDLE Shell 3.13.14
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/rajde/OneDrive/Rajdeep's File/AI/AI_Practical-7/1_AI_Practical-7.py
    Aptitude  Interview         Class
0         85         82      Selected
1         78         75      Selected
2         92         88      Selected
3         60         55  Not Selected
4         72         68  Not Selected
5         88         84      Selected
6         65         60  Not Selected
7         95         90      Selected
8         70         65  Not Selected
9         82         78      Selected
10        58         52  Not Selected
11        90         85      Selected
12        76         72      Selected
13        62         58  Not Selected
14        86         80      Selected
15        68         63  Not Selected
16        80         77      Selected
17        55         50  Not Selected
18        91         89      Selected
19        74         69  Not Selected
Confusion matrix
 [[3 0]
 [0 1]]
Accuracy score:  1.0
Report:
              precision    recall  f1-score   support

Not Selected       1.00      1.00      1.00         3
    Selected       1.00      1.00      1.00         1

    accuracy                           1.00         4
   macro avg       1.00      1.00      1.00         4
weighted avg       1.00      1.00      1.00         4

>>>
Ln: 40   Col: 0
```

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: T101 ARTIFICIAL INTELLIGENCE Practical

Code:- Simple Neural Network for Applicant Prediction

```
#T101 RAJDEEP M PARAB
import pandas as pd
import warnings
warnings.filterwarnings("ignore")
from sklearn.metrics import confusion_matrix, accuracy_score, classification_report
# Reading the applicant dataset
data = pd.read_csv(
    r"C:\Users\rajde\OneDrive\Rajdeep's File\AI\AI_Practical-7\applicant_selection.csv"
)
print(data)
# Splitting the dataset into training and testing data
from sklearn.model_selection import train_test_split
training_set, test_set = train_test_split(
    data,
    test_size=0.2,
    random_state=1
)
# Selecting input features
X_train = training_set.iloc[:, 0:2].values
# Selecting output/class
Y_train = training_set.iloc[:, 2].values
X_test = test_set.iloc[:, 0:2].values
Y_test = test_set.iloc[:, 2].values
# Importing AdaBoost
from sklearn.ensemble import AdaBoostClassifier
# Creating AdaBoost model
adaboost = AdaBoostClassifier(
    n_estimators=100,
    learning_rate=1,
    random_state=1
)
# Training the model
adaboost.fit(X_train, Y_train)
# Predicting test data
Y_pred = adaboost.predict(X_test)
# Adding predictions to test dataset
test_set["Predictions"] = Y_pred
# Creating confusion matrix
results = confusion_matrix(Y_test, Y_pred)
print("Confusion matrix\n", results)
# Calculating accuracy
print("Accuracy score: ", accuracy_score(Y_test, Y_pred))
# Displaying classification report
print("Report: ")
print(classification_report(Y_test, Y_pred))
```

Output:

```
IDLE Shell 3.13.14
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/rajde/OneDrive/Rajdeep's File/AI/AI_Practical-7/2_AI_Practical-7.py
Beginning randomly generated weights:
[[-0.86993568]
 [-0.13949609]
 [-0.56885704]]
Ending weights after training:
[[ 4.80907424]
 [-4.78420777]
 [ 5.11015288]]
Aptitude Score (0 or 1): 1
Interview Score (0 or 1): 1
Communication Score (0 or 1): 1
Considering new applicant:  1 1 1
New output data:
[0.99414752]
>>>
                                                                    Ln: 19  Col: 0
```